Langmuir
Article
pubs.acs.org/Langmuir
© 2010 American Chemical Society

Cleavage-Sensing Redox Peptide Monolayers for the Rapid Measurement of the Proteolytic Activity of Trypsin and α-Thrombin Enzymes

Jocelyne Adjémian,[‡] Agnès Anne,*[,†] Gilles Cauet,[‡] and Christophe Demaille*[,†]

[†]*Laboratoire d'Electrochimie Moléculaire, Unité Mixte de Recherche Université - CNRS No. 7591, Université Paris Diderot, Paris 7, 15 rue Jean-Antoine de Baïf, 75205 Paris Cedex 13, France, and* [‡]*HORIBA Medical Parc Euromédecine, rue du Caducée, BP7290 34184 Montpellier Cedex 4, France*

Received January 27, 2010. Revised Manuscript Received March 10, 2010

Ferrocene (Fc)-labeled peptides are end-grafted onto gold electrodes via a flexible polyethylene glycol (PEG) linker, and their ability to act as substrates for proteolytic enzymes trypsin and α-thrombin is investigated by cyclic voltammetry. It is shown that whereas a short Fc-tetrapeptide substrate is rapidly cleaved by trypsin, a longer Fc-heptapeptide substrate is required for α-thrombin detection. However, in both cases it is observed that not all of the Fc-peptide chains present on the electrode surface are cleavable by the proteases and that the cleavage yield is actually controlled by the surface coverage in the Fc-peptide. Surface dilution of the Fc-peptide using a backfilling molecule such as MCH (6-mercapto-1-hexanol) was required to obtain a cleavage yield larger than 80%. The kinetics of Fc-peptide cleavage by trypsin or α-thrombin is then shown to be adequately described by Michaelis Menten kinetics, allowing enzymatic constants k_{cat} and K_M to be determined. The obtained rate constant values showed that the affinity of the enzymes for their respective Fc-peptide substrates is very high (i.e., low K_M values) whereas that for the cleavage step itself is relatively low (low k_{cat} values). Partial compensation of these parameters yields a fast response of the Fc-peptide electrodes to the proteases in solution in the 1−1000 nM range. The type of molecule used to backfill the Fc-peptide layers, either MCH or PEG_6 chains, is shown to modulate the activity of the proteases versus the Fc-peptide layers: in particular, the PEG_6 diluent is specifically shown to decrease the ability of α-thrombin to cleave its Fc-peptide substrate whereas trypsin activity is unaffected by the presence of PEG chains.

Introduction

Proteases are important physiological enzymes that catalyze the hydrolytic cleavage (proteolysis) of peptide bonds at specific sites along the amino acid sequence, breaking down proteins.[1] By cleaving proteins, proteases are involved in the control of a large number of vital processes such as cell growth, cell death, hemostasis, tissue remodeling, and immune defense. However, the regulation of physiological proteolytic processes, often organized in protease networks that act through activation and subsequent inactivation of proteolytic enzymes, needs to be highly controlled; otherwise, proteases can be dangerous.[2,3] For example, thrombin, a very well studied serine protease, is the main effector protease of the tightly coordinated coagulation cascade. Thrombin generation is closely regulated to achieve locally rapid hemostasis (blood clot formation) after an injury,[4] but excessive or inappropriate proteolytic activity can make this enzyme a pluripotent effector in all pathological thrombotic disorders within the heart and in circulation.[5] Increased proteolytic activity of thombin in blood and tissue has also been found to play a major pathological role in brain diseases such as human central nervous system (CNS) injury[6,7] and Alzheimer's disease.[8,9] Additionally, thrombin has been shown to promote tumor progression and metastasis[10] and has been increasingly proposed as an important biomarker of deregulated programmed cell death (apoptosis).[11,12] Another example in the serine protease family is trypsin, a major pancreatic digestive enzyme responsible for regulating protein degradation.[13] Misbalanced trypsin activity can result in chronic pancreatitis and pancreatic cancers.[14] Elevated levels of the trypsin enzyme may be an aggravating factor in pancreatitis disease and have also been found in a variety of tumors, such as ovarian and colorectal carcinomas.[15] Because the activity of certain proteolytic enzymes is often an indicator of disease states, the development of protease targets, such as protease inhibitors or modulators, is an attractive area for potential new therapeutics, which is being widely explored by pharmaceutical companies.[3,16] In biomedical research and medical disease diagnosis, there continues to be a high demand for convenient methodologies for detecting and measuring the activity of specific proteases in biological samples. The rapid monitoring of the activity of disease-related enzymes (such as thrombin)[17] that are transiently

*Corresponding authors. E-mail: anne@univ-paris-diderot.fr; demaille@univ-paris-diderot.fr.

(1) Barrett, A. J., Rawlings, N. D., Woessner, J. F., Jr., Eds.; *Handbook of Proteolytic Enzymes*, 2nd ed.; Academic: Amsterdam, 2004.
(2) Lopez-Otin, C.; Bond, J. S. *J. Biol. Chem.* **2008**, *283*, 30433–30437.
(3) Turk, B. *Nat. Rev. Drug. Discov.* **2006**, *5*, 785–799.
(4) Crawley, J. T. B.; Zanardelli, S.; Chion, C. K. N. K.; Lane, D. A. *Thromb. Haemost.* **2007**, *5*, 95–101.
(5) Ofosu, F. A. *Thromb Haemost.* **2006**, *96*, 568–577.
(6) (a) Vaughan, P. J.; Pike, C. J.; Cotman, C. W.; Cunningham, D. D. *J. Neurosci.* **1995**, *15*, 5389–5401. (b) Striggow, F.; Riek, M.; Breder, J.; Henrich-Noack, P.; Reymann, K. G.; Reiser, G. *Proc. Natl. Acad. Sci. U.S.A.* **2000**, *97*, 2264–2269.
(7) Chapman, J. *Autoimmun. Rev.* **2006**, *5*, 528–531.
(8) Serruys, P. W.; Vranckx, P.; Allikmets, K. *Int. J. Clin. Pract.* **2006**, *60*, 344–350.
(9) Arai, T.; Miklossy, J.; Klegeris, A.; Guo, J. P.; McGeer, P. L. J. *Neuropathol. Exp. Neurol.* **2006**, *65*, 19–25.
(10) Maragoudakis, M. E.; Tsopanoglou, N. E.; Andriopoulou, P.; Maragoudakis, M. M. *Matrix Biol.* **2000**, *19*, 345–351.
(11) Grabarek, J.; Darzynkiewicz, Z. *Exp. Hematol.* **2002**, *30*, 982–989.
(12) Moffitt, K. L.; Martin, S. L.; Walker, B. *Biochem. Soc. Trans.* **2007**, *35*, 559–560.
(13) Hirota, M.; Ohmuraya, M.; Baba, H. *J. Gastroenterol.* **2006**, *41*, 832–836.
(14) Goldberg, D. M. *Clin. Chim. Acta* **2000**, *291*, 201–221.
(15) Choi, S. Y.; Shin, H. C.; Kim, S. Y; Park, Y. W. *Drug News Perspect.* **2008**, *21*, 417–423.
(16) Law, B.; Tung, C.-H. *Bionconjugate Chem.* **2009**, *20*, 1683–1695.
(17) Edmunds, L. H.; Colman, R. W. *Ann. Thorac. Surg.* **2006**, *82*, 2315–2322.

present in blood or other complex body fluids, in minute amounts, is currently one of the hottest areas of protease research.[3,18]

In recent years, intensive research effort in measuring the activity of proteases in biological environments has highlighted some innovative biosensing devices that exploit a surface-immobilized peptide substrate to signal recognition/cleavage by the protease target. The proteolytic events can be monitored by different methods, but most of the readout methods have essential restrictions, especially for medical diagnostics such as protease blood testing. For instance, whereas optical readout methodologies, in particular, fluorescence resonance energy transfer (FRET), are achieving impressive results, notably in enzymatic activity imaging inside cells with molecular beacon-contrast sensing systems,[19−29] these methods are not readily applicable to achieving sensitive protease assays in complex turbid and colored fluids such as whole blood and plasma. To date, electrochemical detection methods are one of the rare methodological approaches able to meet the ongoing health-care market requirements for molecular diagnostics testing in blood in a sensitive, rapid, cost-effective, compact, easy-to-use format. Applications are exemplified in the established field of blood glucose monitoring[30] and in the emerging area of electrochemical peptide-based sensors for protease activity monitoring.[31−42]

In the practical area of assaying the thrombin activity at the point of care, the shift from optical toward electrochemical sensing has already been taken into account by major companies such as Roche (Switzerland), Abott (U.S.), and Hemosense (U.S.), giving rise to disposable amperometric coagulation rate biosensors on the market, respectively released in their latest versions as Coaguchek XS, I-Stat, and INRatio2 systems. All of these sensing tests use a synthetic peptide labeled with an "electrogenic" group reporter, deposited in dry form on a surface. However, these strip devices have not proved to be a real alternative to centralized laboratory testing for clinical thrombin samples.[43] Such is also the case for the recently disclosed amperometric prototype system from Asulab (Switzerland), whose measurement principle, electrogenic synthetic substrate, and design of the test strips have been the subject matter of several patents;[36] this system has been submitted for first-phase clinical trials.[42] Clearly, an improvement of these existing electrochemical systems in terms of greater reliability, accuracy, and sensitivity is still needed. Actually, it has become important to assess in an early stage of development of many applications for point-of-care testing how the surface chemistry and functionalization will impact the candidate detection technology.

Over the past decade, monolayer-modified electrodes have attracted considerable research interest as well-defined functional surfaces for monitoring enzyme action and activity.[43−49] In a patent issued in April 2007,[37b] Nigretto et al. claimed, in particular, the use of carefully designed self-assembled peptide substrate monolayers to ensure the uniform, selective activity of the immobilized peptide molecules with soluble proteases. However, their new assay format still considers the use of an aromatic amine as the electrochemical reporter, whose sluggish behavior does not allow for easy quantitative characterization of binding events or enzymatic assays in a well-defined heterogeneous format. In the meantime, Liu et al. proposed for the first time a simple, convenient electrochemical heterogeneous format for assaying the activity of a disease-related protease, metalloproteinase MMP7.[38] Their system, also based on a surface end-grafted protease-specific peptide monolayer, now takes advantage of the generic redox ferrocene (Fc) end-labeling strategy. In their working principle studies, a peptide fragment containing the Fc moiety is cleaved and removed from the electrode. The cleavage of the redox-labeling event causes a decreased electrochemical response of the redox-peptide-sensing monolayer electrode depending on the amount of protease in solution, which can be directly analyzed via the sensitive electrochemical voltammetry detection method. To our knowledge, only two additional electrochemical biosensing systems working directly on the principle outlined for the electrochemical MMP7 activity method have been described in the literature, respectively targeted at detecting the activities of cysteine protease caspase 3[40] and the trypsin-like

(18) Giesen, P.; Hemker, H. C.; Al Dieri, R.; Beguin, S. L.; Wagenvoord, R.; Synapse, B. V., The Netherlands; Diagnostic Test for Determining the Concentration of Transient Proteolytic Activity in Composite Biological Media. Eur. Pat. EP 1504269, Nov 13, **2003**.
(19) Salisbury, C. M.; Maly, D. J.; Ellman, J. A. *J. Am. Chem. Soc.* **2002**, *124*, 14868–14870.
(20) Zhu, Q.; Uttamchandani, M.; Li, D. B.; Lesaicherre, M. L. *Org. Lett.* **2003**, *5*, 1257–1260.
(21) Baruch, A.; Jeffery, D. A.; Bogyo, M. *Trends Cell Biol.* **2004**, *14*, 29–35.
(22) Kilian, A. K.; Böcking, T.; Gaus, K.; Gal, M.; Gooding, J. J. *ACS Nano* **2007**, *1*, 355–361.
(23) Grant, S. A.; Weilbaecher, C.; Lichlyter, D. *Sens. Actuators, B* **2007**, *121*, 482–489.
(24) (a) Wegner, G. J.; Wark, A. W.; Lee, H. J.; Codner, E.; Saeki, T.; Fang, S.; Corn, R. M. *Anal. Chem.* **2004**, *76*, 5677–5684. (b) Welser, K.; Grilj, J.; Vauthey, E.; Aylott, J. W.; Chan, W. C. *Chem. Commun.* **2009**, 671–673.
(25) Li, J.; Yeung, E. S. *Anal. Chem.* **2008**, *80*, 8509–8513.
(26) Chen, N.; Zou, J.; Wang, S.; Ye, Y.; Huang, Y.; Gadda, G.; Yang, J. *Biochemistry* **2009**, *48*, 3519–3526.
(27) Zhao, Q.; de Zoysa, R. S. S.; Wang, D.; Jayawardhana, D. A.; Guan, X. *J. Am. Chem. Soc.* **2009**, *131*, 6324–6325.
(28) Orosco, M. M.; Pacholski, C.; Sailor, M. J. *Nat. Nanotechnol.* **2009**, *4*, 257–258.
(29) Salthousea, C. D.; Reynolds, F.; Tama, J. M.; Josephsona, L.; Mahmood, U. *Sens. Actuators, B* **2009**, *138*, 591–597.
(30) Newman, J. D.; Turner, A. P. E. *Biosens. Bioelectron.* **2005**, *20*, 2435–2453.
(31) Abd-Rabboh, H. S. M.; Nevins, S. A.; Dürüst, N.; Meyerhoff, M. E. *Biosens. Bioelectron.* **2003**, *18*, 229–236.
(32) Kitano, H.; Makino, Y.; Kawasaki, H.; Sumi, Y. *Anal. Chem.* **2005**, *77*, 1588–1595.
(33) Ionescu, R. E.; Cosnier, S.; Marks, R. S. *Anal. Chem.* **2006**, *78*, 6327–6331.
(34) Neff, P. A.; Serr, A.; Wunderlich, B. K.; Bausch, A. R. *Chem. Phys. Chem.* **2007**, *8*, 2133–2137.
(35) (a) Mahmoud, K. A.; Kraatz, H.-B. *Chem.—Eur. J.* **2007**, *13*, 5885–5895. (b) Kerman, K.; Mahmoud, K. A.; Kraatz, H.-B. *Chem. Commun.* **2007**, 3829–3831.
(36) (a) Frenkel, E. J.; Haeberli, A.; Moresi, A.; Asulab, C. A., Switzerland; Electrochemical System for Determining Blood Coagulation Time. U.S. Patent 6,352,630, Mar 5, **2002**. (b) Ludin, C.; Wikstroem, P.; Svendsen, L. G.; Schulze, A.; Pentapharm, A. G., Switzerland; Oligopeptide Derivatives for the Electrochemical Measurement of Protease Activity. U.S. Patent 6,495,336, Dec 17, **2002**. (c) Thüerlemann, C.; Haeberli, A.; Frenkel, E. J.; Asulab, C. A., Switzerland; Electrochemical System for the Determination of Blood Coagulation Time System for Differential Determination of a Proteolytic Enzyme Level in a Bodily Fluid. U.S. Pat. Appl. 2007/0009982 A1, Jan 11, **2007**.
(37) (a) Nigretto, J.-M.; Lamblin, G.; Paris, M; Benattar, N. Biopep, France; Method and Device for Measuring an Enzymatic Activity in a Body Fluid. Fr. Patent 00203, **2005**. (b) Nigretto, J.-M.; Lamblin, G.; Paris, M.; Benattar, N. Biopep, France; Method and Device for Measuring an Enzymatic Activity in a Body Fluid. U.S. Patent 0269,854, April 17, **2007**.
(38) Liu, G.; Wang, J.; Wunschel, D. S.; Lin, Y. *J. Am. Chem. Soc.* **2006**, *128*, 12382–12383.
(39) Li, D.; Gill, R.; Freeman, R.; Willner, I. *Chem. Commun.* **2006**, 5027–5029.
(40) Xiao, H.; Liu, L.; Meng, F.; Huang, J.; Li, G. *Anal. Chem.* **2008**, *80*, 5272–5275.
(41) Keiichi, O.; Iwao, M.; Michinori, W.; Shigeori, T. *Anal. Biochem.* **2009**, *385*, 293–299.
(42) Thüerlemann, C.; Haeberli, A.; Alberio, L. *Clin. Chem.* **2009**, *55*, 505–512.
(43) Brock, T. K; Gentile, N. L.; Louie, R. F.; Tran, N. K.; Kitano, T.; Kost, G. J. *Clin. Chem.* **2009**, *55*, 398–399.
(44) Houseman, B. T.; Mrksich, M. *Trends Biotechnol.* **2002**, *20*, 279–28.
(45) Gooding, J. J.; Mearns, F.; Yang, W. R.; Liu, J. Q. *Electroanalysis* **2003**, *15*, 81–96.
(46) Fan, C. H.; Plaxco, K. W.; Heeger, A. J. *Trends Biotechnol.* **2005**, *23*, 186–192.
(47) Di Giusto, D. A.; Wlassoff, W. A.; Gooding, J. J.; Messerle, B. A.; King, G. C. *Nucleic Acids Res.* **2005**, *33*, e64.
(48) Anne, A.; Bonnaudat, C.; Demaille, C.; Wang, K. *J. Am. Chem. Soc.* **2007**, *129*, 2734–2735.
(49) Kerman, K.; Song, H.; Duncan, J. S.; Litchfield, D. W.; Kraatz, H.-B. *Anal. Chem.* **2008**, *80*, 9395–9401.

protease plasmin for the most recent system.[41] The same working principle has also been recently exploited by Willner and coworkers as an analytical tool for probing enzymatic reactions.[39]

We focus here on two other biomedically relevant proteolytic enzymes both in the serine protease family, trypsin and α-thrombin. More than going all the way toward the fabrication of an operational biosensor, the aim of the present work is to bring about insight into the role played by the structure of Fc-peptide monolayers in the ability of such layers to sense the proteolytic activity of enzymes. We experimentally address the issues of (i) the compactness of the Fc-peptide monolayer, (ii) the peptide chain length, and (iii) the nature of the backfilling molecule. We show that even though some general trends regarding the optimization of the sensing capabilities of Fc-peptide layers can be delineated, the structure of the layer has to be finely tuned as a function of the enzyme to be detected.

Experimental Section

Materials. The two ferrocene-labeled peptide disulfide molecules used in this study, Fc-$_{C_{ter}}$[GRPS]$_{N_{ter}}$-PEG disulfide and heptapeptide Fc-$_{C_{ter}}$[RFSRPQL]$_{N_{ter}}$-PEG disulfide, contained a poly(ethylene)glycol PEG_7 disulfide at the N terminus for immobilization on gold surfaces and a ferrocene ethyl unit at the C terminus as a signaling redox tag. These novel peptides were chemically synthesized by Horiba Medical (Montpellier, France) in the framework of a collaboration project aimed at using redox peptide substrates for the electrochemical monitoring of thrombin activity in biological fluids. The ferrocene-peptide disulfides were provided as high-performance liquid chromatography (HPLC) samples whose purity was >95% as confirmed by ESI Q-TOF mass spectroscopy. (See Supporting Information for synthesis and characterization.) Boc(ter-butoxycarbonyl)-[SPR]-pNA peptide, labeled at the C terminus with *p*-nitroaniline unit (pNA), was also a Horiba Medical product and was used as a readily cleavable reference substrate in proteolytic assays in solution (Supporting Information).

The thiol molecules used for backfilling the Fc-peptide monolayers, 6-mercapto-1-hexanol OH-$(CH_2)_6$-SH (hexanethiol MCH) and methoxy-terminated PEG_6 disulfide $(CH_3O\text{-}(CH_2CH_2O)_6\text{-}S\text{-})_2$, were respectively from Sigma-Aldrich (Germany) and Polypure (Oslo, Norway). Stock solutions of MCH and PEG_6 disulfide were made in ethanol at 7 and 4 mM, respectively.

Trypsin (from bovine pancreas, $M_w \approx 23$ kDa, lyophilized) was a Sigma product. A 100 μg/mL solution of trypsin in 1 mM HCl was divided into 10 μL aliquots and stored at −20 °C. Human α-thrombin (50% v/v glycerol/H_2O at 8.8 mg/mL ($M_w \approx 37$ kDa, ~0.24 mM)) was from Haematologic Technologies Inc. (Essex Junction). Serine protease inhibitor 4-(2-aminoethyl)-benzenesulfonyl fluoride (AEBSF) was from Interchim (Montluçon, France).

Other commercial chemicals were reagent grade or better quality and were used as received. All reactions were carried out in polypropylene tubes and protected from light. All aqueous solutions were made with Milli-Q purified water (Millipore).

Cyclic Voltammetry Measurements. Aqueous 1 M $NaClO_4$ was used as the electrolyte solution for all electrochemical studies. Electrochemical experiments were performed with a conventional three-electrode configuration consisting of a gold disk working electrode, a platinum wire counter electrode, and a reference electrode consisting of a KCl saturated calomel electrode (SCE). The SCE reference electrode was separated from the supporting electrolyte solution with a bridge terminated with a glass frit containing an aqueous solution of 1 M NaCl. The peak potentials were measured to an accuracy of ±5 mV. For the electrochemical pretreatment of the gold electrodes, the bridge contained 1 N H_2SO_4. All electrochemical experiments were performed using conventional instrumentation.[50] Cyclic voltammograms were recorded without ohmic drop compensation. All potentials are reported versus SCE. The temperature in the electrochemical experiments was 20 °C.

Pretreatment of the Polycrystalline Gold Electrodes. Gold disk working electrodes were constructed in-house by sealing lengths of gold wire (99.99%, Goodfellow, 0.5 mm diameter) within polypropylene bodies. For all experiments, the electrodes were polished to a mirror finish using progressively finer grades of alumina polishing suspensions (3, 0.5, and 0.05 μm, Buehler), followed by ultrasonication in water and ethanol. Prior to the preparation of Fc-peptide-PEG monolayers, the freshly polished electrodes were electrochemically cleaned by cyclic voltammetry at a scan rate of 0.2 V/s in 1 N H_2SO_4 as reported previously.[50] The final electrochemical oxidation step was followed by electrochemical reduction of the gold oxide monolayer via a reverse potential scan down to +0.2 V. The effective areas of the electrodes S_{eff} were derived from the charge associated with the gold oxide reduction peak. The accepted average value of 400 μC/cm^2 for a gold oxide monolayer on polycrystalline gold[51] was used in this work for the estimation of S_{eff}, the effective electrode surface area. We found S_{eff} to be (6 to 11) × 10^{-3} cm^2, which typically correspond to a roughness factor of 3−5. The thus-pretreated gold electrodes were quickly rinsed with water and ethanol and then immediately used for reaction with the Fc-peptide disulfide substrate.

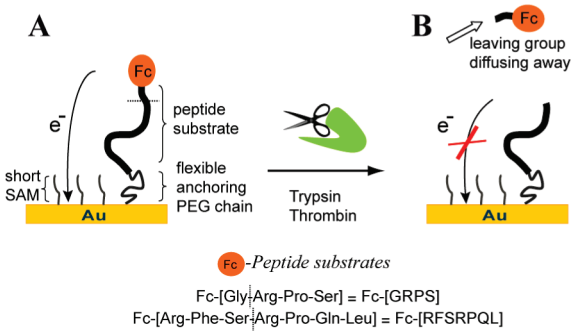
Preparation of Mixed Fc-Peptide Monolayers in a Rapid Two-Step Assembly. The mixed Fc-peptide monolayers on gold electrode surfaces were prepared and characterized according to a rapid two-step assembly procedure as follows: In the first step, the freshly prepared gold electrode surface was incubated at ambient temperature in 100 μL of a low-concentration solution (~2 to 5 μM) of the Fc-peptide disulfide in ethanol for an appropriate very short time, depending on the final chain-grafting coverage sought. The Fc-peptide-modified gold electrode was rinsed with ethanol and then water and immediately analyzed by cyclic voltammetry in 1 M $NaClO_4$ (at 100 V/s) for a rough estimation of the Fc-peptide coverage. A short reaction time ranging from 30 to 120 s generally yielded a final surface coverage of ~0.4 × 10^{-11} to 2 × 10^{-11} mol/cm^2. However, variations in the Fc-peptide coverage depending on the type of gold surface have been observed. Longer immersion times allowed us to prepare more densely grafted Fc-peptide layers approaching the maximum surface coverage (saturation of ca. 6 × 10^{-11} mol/cm^2). The Fc-peptide-modified electrode was then briefly washed with water and ethanol before immediate backfilling. In this second reaction step, the electrode was incubated in 7 mM ethanolic MCH or 4 mM PEG_6 disulfide diluent solution for ca. 1 to 2 h to displace any nonspecifically adsorbed Fc-peptide and backfill the remaining electrode area. Prior to CV analysis, the thus-mixed Fc-peptide monolayer electrode was immersed in ethanol for 5 min to eliminate any residual alkane thiols or PEG disulfide and then rinsed with water. After CV analysis, the Fc-peptide electrode was submitted for enzymatic assays. Otherwise, the electrode was kept wet at all times under dearated aqueous or ethanolic conditions for prolonged stability. Under these conditions, the modified electrodes were stable for at least 2 days. Details regarding the CV characterization of the assembly process can be found in the Supporting Information.

Surface Enzymatic Assay Conditions. The assay buffer consisted of 50 mM Tris−HCl (pH 8.4) and 20 mM $CaCl_2$ for trypsin and 50 mM Tris−HCl (pH 8.4) and 100 mM NaCl for thrombin. The surface reaction of protease was carried out in a test tube maintained at 37 °C containing 100 μL of the assay buffer and a given amount of protease, preactivated for 2 min. The Fc-peptide-modified electrode was then immersed in the protease solution for a given time period. The thus-reacted electrode was

(50) Anne, A.; Demaille, C.; Moiroux, J. *Macromolecules* **2002**, *35*, 5578–5586.

(51) (a) Michri, A. A.; Pshenichnikov, A. G.; Burshtein, R. *Kh. Elecktrokhim.* **1972**, *8*, 364–366. (b) Rand, D. A. J.; Woods, R. J. *Electroanal. Chem.* **1972**, *35*, 209–218.

Scheme 1. Schematic Representation of the Proposed Electrochemical System for the Detection of Trypsin and Thrombin Activity[a]



[a] (A) The initial system consisted of a diluted self-assembled (SAM) redox-active ferrocene (Fc)-peptide substrate monolayer on a gold surface. (B) Cleavage of the Fc-peptide system by the protease enzyme (scissors) leads to the separation of the electroactive reporter ferrocene, which diffuses away from the modified-electrode surface. (---) Proteolytic cleavage site.

immediately rinsed with cold water to stop the reaction, followed by rapid immersion in aqueous 1 M $NaClO_4$ before analysis by CV for Fc-tag signal loss.

Results and Discussion

Rational Design of the Protease-Sensing Monolayer: Electrochemical Detection Principle. The structure of the protease-sensing monolayer developed here is schematically presented in Scheme 1; it is rationally designed following the guidelines discussed below.

A short peptide (four or seven amino acids long) is labeled at its C_{ter} extremity by an electrochemically detectable ferrocene (Fc) group and is attached to the surface of a planar gold electrode by its N_{ter} extremity via a thiol−gold bond. The thiol group is borne by a short PEG_7 (poly(ethylene glycol)) spacer chain (termed PEG in the following text) that is expected to add more flexibility to the end-anchored Fc-peptide system. A low-potential alkyl ferrocene (Fc) label was specifically chosen because it is a redox molecule having simple electrochemical behavior that is quite insensitive to the surrounding medium. The sequence of the peptide is chosen so that the chain can act as a substrate for the target protease (here trypsin and thrombin). In other words, the Fc-peptide chains are chosen so that they can be readily cleaved by these later protease enzymes. The sequence should therefore contain at least an argininyl residue (R) because trypsin-like enzymes specifically cleave peptide chains on the carboxyl side of this residue. However, thrombin is a more discriminating protease than trypsin because of extra-surface structures that influence interactions with macromolecular substrates.[52] Actually, it is well recognized that thrombin has a marked specificity preference for proline (P) located at the 2 position that is N terminal to the cleavage site.[53] Therefore, for optimized cleavage efficiency, the synthetic peptide substrate is designed to contain the (P-R) motif. The Fc-peptide-PEG monolayer is then backfilled by short diluent molecules, which can be either short hydrophilic thiols such as MCH (6-mercapto-1-hexanol) or PEG_6 chains. The primary role of these surface diluent molecules is to prevent the nonspecific adsorption of the peptide chain onto the gold electrode surface. Such an unwanted adsorption would result in a flat-lying peptide chain that would then become inaccessible to the enzymes. Because proteases are, in fine, to be assayed in biological fluids, the backfilling molecules should also prevent the nonspecific adsorption of proteins, which could passivate the sensing layer by blocking the access to the surface-grafted peptide substrate.

The Fc label of the peptide chains can be conveniently detected by cyclic voltammetry: upon linearly scanning the electrode potential E as a function of time toward oxidative potentials, one can trigger the single-electron oxidation of the ferrocene head by reversibly oxidizing it (Scheme 1A). Oxidation of the Fc head gives rise to a current passing through the electrode, which is used as a detection signal for the labeled peptide chain. The low potential of the alkyl ferrocene head avoids the detection of interfering electroactive molecules that may be present in the enzyme-containing fluid to assay. An analysis of the current signal allows the number of labeled peptide chains present on the electrode to be quantified.

The electrochemical detection strategy of protease activity consists of monitoring the loss of the ferrocene label resulting from the cleavage of the peptide chain by the proteolytic enzyme to be assayed (Scheme 1). The configuration implemented here therefore belongs to the "signal off" class of biosensors because the enzymatic activity results in the disappearance of the electrochemical signal. Measuring the current signal loss for given immersion times of the sensing electrode in the enzyme-containing solution should make it possible to quantify the enzyme concentration.

Proteolytic Activity of Trypsin toward MCH-Diluted Fc-[GRPS]-PEG Layers. The peptide chain of the first Fc-peptide-PEG molecule used in the present work is composed of four amino acids, namely, $_{N_{ter}}$[serine (S)-proline (P)-arginine (R)-glycine (G)]$_{C_{ter}}$. Rapid cleavage of this Fc-[GRPS]-PEG molecule in solution by both trypsin and thrombin was verified by high-performance liquid chromatography (HPLC) (Supporting Information). Mixed monolayers composed of the Fc-tetrapetide and 6-mercapto-1-hexanol (MCH) were assembled onto gold electrode surfaces following a rapid two-step procedure as described in the Experimental Section. The Fc-[GRPS]-PEG-modified gold electrode was then characterized by cyclic voltammetry in 1 M $NaClO_4$ (at 100 V/s). As seen in Figure 1A and B (black traces), dilute Fc-[GRPS]-PEG/MCH layers give rise to almost ideal surface signals typical of surface-confined species undergoing Nerstian electron transfer:[54] the peak current is proportional to the scan rate v, the peak-to-peak separation is low (ca. 10 mV), and importantly, the half sum of the forward and backward peaks is ~155 mV/SCE, which is quite close to the standard potential of the Fc heads of the Fc-[GRPS]-PEG chains in solution (150 mV/SCE). Equally important is the fact that the width at midheight of the forward or backward peaks is ~105 mV, which is close to the value of 90 mV expected for noninteracting surface-bond redox species located outside the double layer. Integration of the forward or backward peaks of the voltammogram yields a surface concentration of Fc-[GRPS]-PEG chains, Γ_0, that was optimized to $\Gamma_0 \leq 1 \times 10^{-11}$ mol/cm^2 (Supporting Information). As a benefit of the relatively high scan rate used (100 V/s) and the presence of MCH backfilling, the background current is low and the peak-shaped Faradaic signal is clearly observed even for low Fc-[GRPS]-PEG coverages approaching 10^{-12} mol/cm^2 (Figure S1).

When these Fc-[GRPS]-PEG/MCH layers were immersed in a 100 μg/mL trypsin assay solution, their response was observed to be remarkably fast because, as seen in Figure 1A, a 75% decrease

(52) Bode, W.; Turk, D.; Karshikov, A. *Protein Sci.* **1992**, *1*, 426–471.
(53) Backes, B. J.; Harris, J. L.; Leonetti, F.; Craik, C S.; Ellman, J. A. *Nat. Biotechnol.* **2000**, *18*, 187–193.
(54) Laviron, E. *J. Electroanal. Chem.* **1979**, *101*, 19–28.

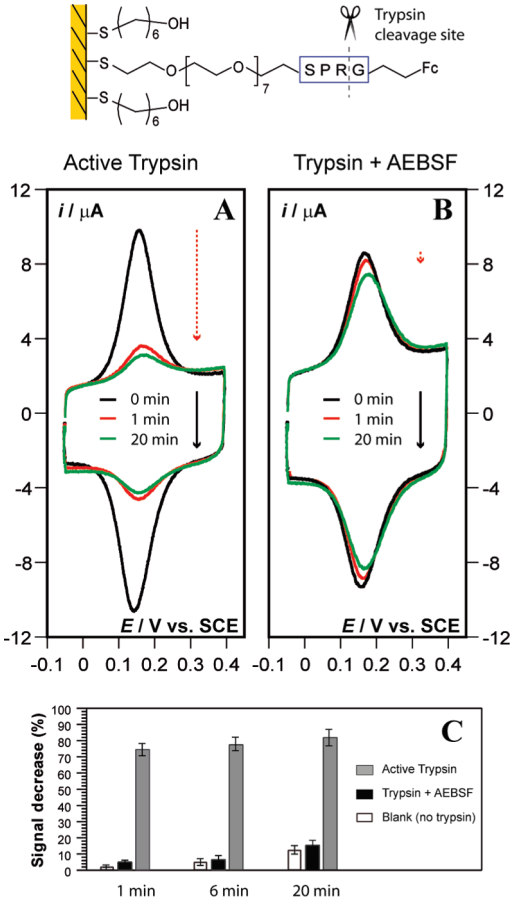


Figure 1. Cyclic voltammetry at a gold electrode bearing a diluted Fc-[GRPS]-PEG/MCH layer (black traces). Proteolytic activity of trypsin toward diluted Fc-[GRPS]-PEG/MCH peptide layers. (A, B) Typical cyclic voltammetry responses of the Fc-peptide layers after immersion times of 1 and 20 min at 37 °C in 100 μg/mL trypsin protease solutions. (A) Active trypsin: a 75% CV signal decrease is effected in 1 min, as illustrated by the dashed red arrow. (B) Trypsin deactivated with serine protease inhibitor AEBSF: only a small (ca. 5%) CV signal decrease is observed after 1 min, as illustrated by the dashed red arrow. (C) Comparison between the CV signal decreases (percentage) obtained at immersion times of 1, 6, and 20 min with active trypsin, inhibited trypsin (trypsin + AEBSF), or the absence of the enzyme (blank). Error bars are the standard deviation of at least three repetitive experiments. Inhibition conditions with AEBSF: trypsin was deactivated in trypsin protease assay buffer (50 mM Tris, 20 mM $CaCl_2$, pH 8.4) for 30 min at 37 °C with 1 mM AEBSF. CVs of electrodes bearing before-immersion redox peptide surface coverages Γ_0 of (A) $\sim 9 \times 10^{-12}$ and (B) $\sim 5 \times 10^{-12}$ mol/cm^2 are shown. CV analysis was conducted at a scan rate of 100 V/s in 1 M $NaClO_4$ supporting electrolyte at 20 °C.

in signal is obtained in 1 min and is attributed to the rapid cleavage of the peptide by trypsin. (See below.) This response time is at least 10 times faster than those reported for electrochemical protease activity assay systems using end-anchored peptide layers (response time of at least 20−30 min).[38−41] We attribute such a spectacular improvement to the structure of the end-grafted protease substrate molecule designed for the present work that uniquely integrates a flexible (PEG) anchor, probably facilitating the access of the protease to the peptide cleavage site.

Incubating the electrode in the trypsin assay solution for a duration longer than a few minutes (e.g., up to 20 min) results in a modest further overall decrease in the signal to 80%. The specificity of the trypsin-catalyzed cleavage of the end-grafted peptide was ascertained by immersing a dilute Fc-[GRPS]-PEG/MCH-layer-modified electrode in a solution containing trypsin treated with irreversible serine protease inhibitor AEBSF. As expected, after 1 min of immersion in the deactivated trypsin solution, only a very weak (less than 5%) signal decrease was measured (red trace in Figure 1B). This very small yet nonzero signal decrease can be attributed to the residual activity of trypsin deactivated in 1 mM AEBSF for 30 min (Supporting Information).

Control experiments were carried out by monitoring the signal decreases (%) observed when Fc-[GRPS]-PEG/MCH-modified electrodes were immersed in an assay buffer solution containing either active trypsin, AEBSF-inactivated tryspin, or no enzyme (blank solution). The results are compared in the bar graph shown in Figure 1C for immersion times of 1, 6, and 20 min. In the absence of the enzyme, only a modest ~5% signal decrease is observed after 6 min and a 15−20% decrease is observed after ~20 min. This time-dependent loss of signal is attributed to the spontaneous detachment of a small fraction of the Fc-[GRPS]-PEG chains from the electrode surface at 37 °C. Chemical degradation of the Fc moiety during surface enzymatic assays can be excluded as an alternative explanation of signal loss because control HPLC experiments, carried out under the same conditions as surface assays, did not show any sign of loss of the Fc moiety from the Fc-peptide over a 1 h period. In the inactivated trypsin solution, for immersion times >1 min, the signal decrease is barely larger than the one observed using a blank solution. Importantly, as one can see from Figure 1C, after 1 min of immersion the signal decrease due to the catalytic action of active trypsin is much larger than the one observed for both AEBSF inactivated trypsin and the blank. Such is still largely the case for 6 min of immersion. For 20 min of immersion, the chain loss in the blank solution and the residual activity of the deactivated trypsin somewhat blur the picture because they can account for a nonnegligible fraction of the signal decrease. Thus, it can be concluded that the best sensitivity for assaying the specific activity of trypsin is achieved if an immersion time of up to 6 min is used.

Effect of the Surface Coverage in Fc-[GRPS]-PEG Chains on the Cleavage Efficiency of Trypsin. The effect of the surface coverage in Fc-[GPRS]-PEG chains on the ability of trypsin to cleave the peptide chain was systematically investigated. Mixed MCH/Fc-peptide layers of various surface coverages Γ_0 were assembled as described above. The electrodes were then immersed in the trypsin assay solution at [trypsin] = 100 μg/mL for a long enough time (20 min) to compensate for the likely dependence of the cleavage kinetics on Γ_0 and to focus on the maximum cleavage yield versus Γ_0 relationship. The resulting CV signal decrease (%) is plotted as a function of the initial Fc-peptide coverage Γ_0 in Figure 2. One can see that the signal decrease ranges from a low value of 50% for undiluted compact layers ($\Gamma_0 = 6 \times 10^{-11}$ mol/cm^2) and increases as the Fc-[GRPS]-PEG molecules are surface diluted by MCH to reach a high of 85 ± 5% for Γ_0 values lower than 10^{-11} mol/cm^2 .

The observed dependence of the cleavage efficiency as a function of the Fc-[GRPS]-PEG chain coverage demonstrates that if very responsive Fc-peptide layers are to be assembled then enough room has to be left around each end-anchored Fc-peptide for the protease to cleave the peptide chain. We also consistently observed an ~+10−15 mV shift in the potential of the CV peaks after the electrode was placed in contact with the enzyme whereas the peak-to-peak separation remained unchanged, even for a low-cleavage yield. This shift can be attributed to a modification of the microenvironment of the Fc head possibly resulting from the

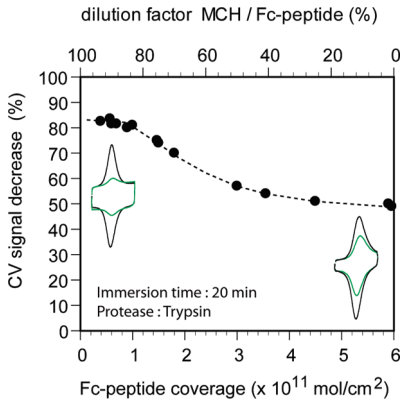


Figure 2. Mixed Fc-[GRPS]-PEG/MCH peptide monolayers on gold surfaces for the detection of the proteolytic activity of trypsin. Relationship plot between the initial Fc-peptide surface coverage Γ_0 and the decrease in the CV signal of the redox peptide layer response after 20 min of immersion at 37 °C in a 100 μg/mL active trypsin solution. CV thumbnails are given to illustrate the relationship between the Fc-peptide coverage and CV signal decrease. Each value is an average of at least two independent determinations with freshly prepared electrodes. See the Experimental Section for exposure times to the peptide substrate and the concentrations of MCH considered in preparation of the mixed Fc-peptide/MCH electrodes. The dashed line is meant to be a visual guide. Experimental enzymatic assay buffer and CV analysis conditions are the same as in Figure 1.

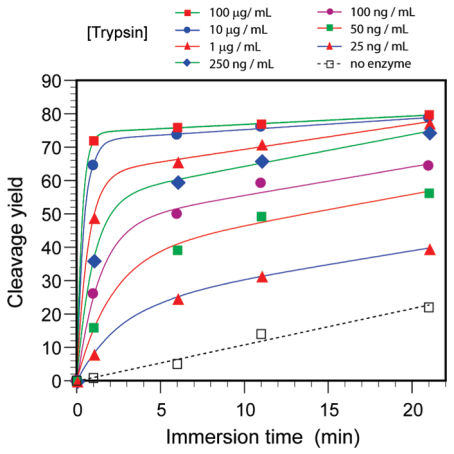


Figure 3. Proteolytic trypsin activity detection using Fc-[GRPS]-PEG/MCH layers in a protease buffer assay. The relationship plots between immersion time and cleavage yield (θ_c) result from active trypsin at different concentrations varying from 25 ng/mL to 100 μg/mL. The solid lines represent visual guides. Experimental conditions are the same as in Figure 1. Several electrodes displaying an Fc-[GRPS]-PEG chain coverage in the range of $(0.5-1) \times 10^{-11}$ mol/cm^2 were used to acquire the data.

weak adsorption of enzyme molecules in the Fc-peptide layer. Interestingly, the fact that at high chain coverage a significant fraction of the Fc-peptide chains do not seem to be cleavable by trypsin may indicate that, for dense layers, the Fc-peptide-PEG-disulfide molecules form compact domains on the surface in which the peptide chains are not accessible to the enzyme, as also observed by Hardesty et al.[55]

However, for low enough chain coverage ($\Gamma_0 \leq 10^{-11}$ mol/cm^2) the Fc-peptide chains behave as freely accessible isolated chains because they are then readily cleavable. We can thus further characterize the sensing capability of these optimized dilute layers in terms of sensitivity.

Trypsin Concentration and Immersion Time Dependence of the Electrochemical Response of Fc-[GRPS]-PEG/MCH-Modified Electrodes. The ability of the MCH-diluted Fc-[GRPS]-PEG-monolayer-modified electrodes to detect low concentrations of trypsin is considered below. The complete immersion time dependence of the response of the Fc-peptide electrodes for various concentrations of trypsin, ranging from 100 μg/mL down to 25 ng/mL, is presented in Figure 3 in terms of the Fc-peptide cleavage yield θ_c. θ_c was calculated by $\theta_c = 100(1 - (\Gamma/\Gamma_0))$, where Γ_0 and Γ stand for the Fc-peptide coverage measured before and after immersion in the protease solution, respectively.

One can see that, no matter the trypsin concentration, the cleavage yield increases initially very rapidly, in less than 10 min, before evolving much more slowly at longer time. It can also be seen that the specific trypsin-catalyzed peptide cleavage can be clearly differentiated from spontaneous chain loss (blank solution, no enzyme) for trypsin concentrations as low as 25 ng/mL. In fact and as discussed above, this is particularly the case for the data collected after short immersion times in the trypsin solution. This is illustrated in Figure 4 where the cleavage yields recorded for 1, 6, and 11 min immersion times are plotted as a function of the trypsin concentration.

One can see from the thus-obtained calibration curves that the variation of the cleavage yield with the immersion time can be used to assay trypsin accurately from a concentration of 100 μg/mL down to 25 ng/mL (i.e ~1 μM down to ~1 nM). This range is relevant to the detection of subnormal trypsin levels indicative of pancreatic disorders.[57]

The calibration curves corresponding to an immersion time of 6 or 11 min may be preferably used to assay low trypsin concentrations (< 100 ng/mL) because they yield a larger signal decrease than at 1 min of immersion time. For higher trypsin concentrations (> 100 ng/mL), it may be advantageous to use the working curve corresponding to 1 min of immersion time because for longer immersion times the calibration curves tend to display saturating behavior (i.e., the cleavage yield is almost at its maximum within ~5 min, Figure 4A). However, provided accurate measurements of the signal decrease are made, one can also simply use the working curve recorded at 1 min for the whole range of trypsin concentration explored because within this range the cleavage yield measured at 1 min markedly depends on the trypsin concentration.

Kinetics of the Trypsin-Catalyzed Proteolysis of End-Grafted Fc-[GRPS]-PEG Chain: Kinetic Model. Hydrolysis of the surface-bound Fc-peptide chains by trypsin can be kinetically represented by the following two-step reaction sequence, in agreement with the known mechanism of action of serine protease:[58]

$$E + S_{surf} \underset{k_d}{\overset{k_a}{\rightleftharpoons}} (ES)_{surf} \xrightarrow{k_{cat}} E + P_1 + P_{2,surf}$$

First, molecular recognition of the surface-grafted Fc-peptide substrate (S_{surf}) by the protease enzyme (E) present in solution yields the surface-bound enzyme/substrate complex $(ES)_{surf}$.

(55) Hardesty, J. O.; Cascão-Pereira, L.; Kellis, J. T.; Robertson, C. R.; Frank, C. W. *Langmuir* **2008**, *24*, 13944–13956.

(56) http://www.ria-cis.com/ressources/produits/02_GB_RgTrypsin_Mod10.pdf; Cisbio Bioassays RIA-gnost trypsin kit.

(57) Millington, R. B.; Mayes, A. G.; Blyth, J.; Lowe, C. R. *Sens. Actuators, B* **1996**, *33*, 55–59.

(58) Hedstrom, L. *Chem. Rev.* **2002**, *102*, 4501–4523.

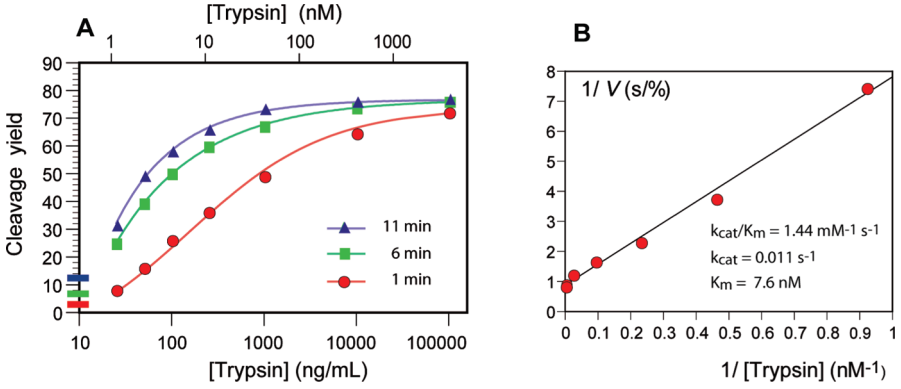


Figure 4. Proteolytic trypsin activity detection using Fc-[GRPS]-PEG/MCH layers. (A) Calibration curves corresponding to immersion times of 1, 6, and 11 min in active trypsin assay solutions at 37 °C. The thick lines crossing the bottom of the "cleavage yield" axis are blank data for [trypsin] = 0 nM corresponding to immersion times of 1 (red), 6 (green), and 11 min (blue). The investigated dynamic range covers the serum-relevant trypsin concentrations for healthy subjects, 140−400 ng/mL, and also explores concentrations below and above the limits of normality.[56] The solid-line curves are meant to be visual guides. (B) Inverse plot of the initial velocity V of proteolytic cleavage by trypsin as measured in terms of the variation of the cleavage yield over 60 s versus trypsin concentration (data shown in part A for the 1 min assays). The solid line represents the linear regression of the data. The Michaelis−Menten kinetic constants derived from this line, using eq 1, are shown.

Second, the complex dissociates to release in solution both the protease enzyme and the cleaved Fc-tagged C_{ter} part of the peptide (P_1) while leaving on the surface the redox-inactive N_{ter} part of the peptide ($P_{2,surf}$).

Release of the fragments produced by cleavage of the peptide substrates by serine protease enzymes is known to occur in two sequential first-order steps but is represented here by a global first-order reaction (rate constant k_{cat}) for the sake of simplicity. For the same reason, we assume that the enzyme concentration in the vicinity of the electrode is constant and equals the initial enzyme concentration C^0_E. This implies that mass transport of the enzyme toward the Fc-peptide-bearing electrode does not limit the rate of the surface enzymatic reaction. This assumption may lead to an underestimation of the actual rate of surface reactions.[59] Following Gutierrez et al.,[60] the initial enzymatic rate, V_0, can then be approximated by

$$V_0 = -\left.\frac{\partial \Gamma}{\partial t}\right|_{t=0} \approx \frac{k_{cat}\Gamma_0 C^0_E}{K_M + C^0_E} \quad (1)$$

with Γ and Γ_0 being the instantaneous and initial surface concentrations in peptide-borne Fc heads, respectively, and K_M being the Michaelis constant defined as $K_M = (k_d + k_{cat})/k_a$.

The initial velocity V of proteolytic cleavage by trypsin can be expressed in terms of the variation of the cleavage yield θ_c provided that the evolution of the electrochemical signal is specific to the cleavage of the peptide by the enzyme (which is the case for short immersion times), yielding

$$V = \left.\frac{\partial \theta_c}{\partial t}\right|_{t=0} = 100 V_0/\Gamma_0 = 100\frac{k_{cat}C^0_E}{K_M + C^0_E} \text{ or } 1/V$$
$$= \frac{1}{100}\left(\frac{K_M}{k_{cat}} \times \frac{1}{C^0_E} + \frac{1}{k_{cat}}\right) \quad (2)$$

We approximate V by the variation of the cleavage yield over 60 s (i.e., we take $V \approx \theta_c(t = 1\text{ min})/60$ (in % per second)). From the time evolution of θ_c presented in Figure 3, it can be seen that this approximation is reasonable, even though it can lead to an underestimation of the initial cleavage rate especially at high enzyme concentration.

Figure 4B shows the variation of the thus-derived V values for the various trypsin concentrations explored in the form of a $1/V$ versus $1/C^0_E$ Lineweaver−Burk plot. The observed linear variation of $1/V$ as a function of $1/C^0_E$ indicates that, at least in its initial stage, cleavage of the surface-grafted Fc-peptide by trypsin can be kinetically described by the above given Michaelis−Menten formalism. This linear variation also confirms the proportionality between V_0 and Γ_0 predicted by eq 1 because one should recall that several electrodes were used to acquire the data presented in Figure 4 (at least one electrode per data point) and that the individual Γ_0 values of each of these electrodes were used to derive θ_c.

Moreover, as a benefit of the large range of trypsin concentration explored and the well-defined Lineweaver−Burk plot thus obtained, the values of both the k_{cat} and k_{cat}/K_M constants can be derived from linear regression analysis of the data presented in Figure 4B. Using eq 2, the following values are thus obtained: $k_{cat} = (1.1 \pm 0.2) \times 10^{-2}\ s^{-1}$ and $k_{cat}/K_M = (1.44 \pm 0.1) \times 10^6\ M^{-1}\ s^{-1}$.

The value derived for second-order constant k_{cat}/K_M is relatively high because it falls in the range of values typically measured for the trypsin-catalyzed cleavage of peptides in solution (10^6−$10^7\ M^{-1}\ s^{-1}$).[58] However, we notice that the value we obtained for k_{cat} is relatively low considering that k_{cat} for the hydrolysis of peptides by serine proteases in solution can be as high as a few hundred reciprocal seconds.[58] It ensues that, from the large value of k_{cat}/K_M determined above, we derive a remarkably low value for K_M: $K_M = (7.6 \pm 2)$ nM. Considering the definition of $K_M = (k_d + k_{cat})/k_a$, it is clear that the present situation, where both k_{cat} and K_M are low, indicates that the affinity of the enzyme for the grafted Fc-peptide chain is very high whereas the cleavage step itself is slow. Nevertheless, the combination of both factors results in a fast, sensitive response of the Fc-peptide-PEG sensing layer to the presence of the enzyme.

Having optimized the structure of the sensing Fc-[GRPS]-PEG/MCH monolayer for rapid, high-yield cleavage by trypsin, it is relevant to wonder if such a layer can also be used to detect another clinically relevant enzyme of the serine protease family: thrombin.

Proteolytic Activity of α-Thrombin Using Fc-peptide-PEG/MCH Layers: Effect of the Chemical Nature of the Sensing Peptide on the Protease Kinetics. As seen in Figure 5A, when a MCH/Fc-[GRPS]-PEG monolayer bearing

(59) Bourdillon, C.; Demaille, C.; Moiroux, J.; Saveant, J.-M. *J. Am. Chem. Soc.* **1999**, *121*, 2401–2408.
(60) Gutiérrez, O. A.; Chavez, M; Lissi, E. *Anal. Chem.* **2004**, *76*, 2664–2668.

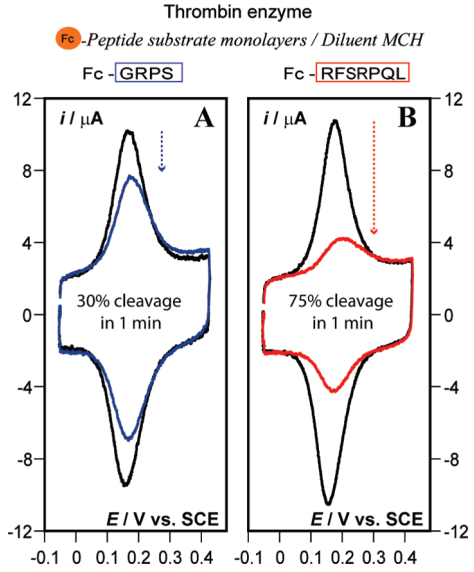


Figure 5. Performances of redox Fc-peptide-PEG/MCH layers on gold for the detection of proteolytic thrombin activity as a function of the nature of the sensing peptide substrate after 1 min of immersion at 37 °C in a 100 μg/mL active thrombin solution. (A) Shown are the CV layer response for the [GRPS] tetrapeptide sensing substrate. A 30% signal decrease is effected by thrombin in 1 min as indicated by the dashed blue arrow. (B) CV layer response for the [RFSRPQL] heptapeptide sensing substrate. An optimized 75% signal decrease is effected by thrombin in 1 min as indicated by the dashed red arrow. Thrombin assay buffer: 50 mM Tris buffer (pH 8.4, 0.1 M NaCl). CV analysis: scan rate v = 100 V/s in 1 M $NaClO_4$ supporting aqueous electrolyte at 20 °C. Redox surface coverages before immersion Γ_0: (A) ~8.9×10^{-12} and (B) ~8.5×10^{-12} mol/cm^2. Effective surfaces S_{eff}: (A) ~9.4×10^{-3} and (B) ~10.7×10^{-3} cm^{-2}.

a gold electrode was immersed in a 100 μg/mL thrombin assay solution (50 mM Tris buffer at pH 8.4, 0.1 M NaCl) for 1 min, a modest 30% signal decrease was noted.

This signal loss is quite small as compared to the 75% signal decrease observed when the same kind of electrode was used to assay trypsin activity under the same conditions. (See above.) Considering the fact that thrombin is only slightly larger ($M_w \approx$ 37 kDa) than trypsin ($M_w \approx$ 23 kDa), steric factors cannot be evoked to explain the low cleavage yield of the Fc-peptide by thrombin. Moreover, we verified that lowering the peptide coverage Γ_0 to below 10^{-11} mol/cm^2 (down to 2×10^{-12} mol/cm^2) did not significantly increase this cleavage yield. Another factor that may control the response of the Fc-[GRPS]-PEG/MCH-modified electrode is the intrinsic reactivity of the protease enzyme toward the specific sequence of the peptide. In our ongoing research effort to design effective thrombin-cleavable Fc-peptide substrates with increased selectivity in solution, we synthesized many Fc-peptides whose length and sequence were chosen according to the following guidelines. Structural studies of α-thrombin and its natural substrates,[61,62] combined with screening tests,[62−67] have shown that optimum heptapeptide substrates for this enzyme have the sequence $_{N_{ter}}$[(F, L, V or Nle)-(Q or V, L, I)-P-R-S-F-(R or K)]$_{C_{ter}}$, with the cleavage site being the R−S bond. Screening tests that we performed in solution showed that, among these peptides, thrombin displayed the highest activity toward the peptide sequence leucine (L), glutamine (Q), proline (P), arginine (R), serine (S), phenylalanine (F), and arginine (R) (i.e., $_{N_{ter}}$[L-Q-P-R-S-F-R]$_{C_{ter}}$). We therefore selected this sequence, synthesized the corresponding Fc-$_{C_{ter}}$[RFSRPQL]$_{N_{ter}}$ peptide disulfide, and used it as described above to construct a dilute MCH/Fc-peptide monolayer on the surface of a gold electrode.

As one can see from Figure 5B, when the Fc-[RFSRPQL]-PEG-bearing electrode was immersed in a 100 μg/mL thrombin solution for 1 min a cleavage yield of 75% was obtained; this is much larger than the one that we recorded when the end-grafted Fc-[GRPS]-PEG peptide was used as a substrate for thrombin (Figure 5A). This difference in cleavage yields reflects the increased reactivity of the thrombin enzyme for the Fc-[RFSRPQL] peptide as compared to that of the Fc-[GRPS] peptide. Incidentally, this result demonstrates that, provided Fc-peptides are immobilized in optimized monolayers onto electrodes, the electrochemical response of surface-grafted Fc-peptide layers can reliably be used to evaluate the intrinsic reactivity of proteases versus specific peptide substrates.

The capability of Fc-[RFSRPQL]-PEG-bearing electrodes to assay thrombin was further evaluated by measuring the decrease in the Fc head coverage (i.e., the cleavage yield) after 1 min of immersion time in a thrombin solution. As seen from the resulting calibration curve reproduced in Figure 6, measuring the decrease in CV signal at these modified electrodes allows the concentration of active thrombin to be accurately measured for an enzyme concentration in the 25 ng/mL to 100 μg/mL range in a very short time. To control bleeding, the thrombin concentration is generally approximately 25 ng/mL to 5 μg/mL (~100 to ~20 000 U/mL).[68] We verified that a 1 min immersion of the Fc-[RFSRPQL]-PEG-bearing electrode in an assay solution containing no thrombin yielded a negligibly small signal decrease. The specificity of thrombin detection was ascertained by observing that when thrombin was inactivated by AEBSF only the residual activity of the deactivated enzyme was recorded (Figure S4) .

Kinetics of the Cleavage of End-Grafted Fc-[RFSRPQL]-PEG Chains by Thrombin. Proceeding as described above for the case of the trypsin-catalyzed cleavage of the Fc-[GRPS]-PEG layer, an approximate value of the initial velocity of the cleavage of Fc-[RFSRPQL]-PEG chains by thrombin is obtained from the variation of the cleavage yield over 60 s of immersion time. The data presented in Figure 6 were thus used to build the $1/V$ versus $1/C^0_E$ plot presented in the inset of Figure 6. A linear variation is observed, indicating that the kinetics of the thrombin-catalyzed Fc-[RFSRPQL] cleavage is of the Michaelis−Menten type. From this variation, the following best-fit values for the kinetic constants are obtained: $k_{cat} = (1.0 \pm 0.2) \times 10^{-2}$ s^{-1}, $k_{cat}/K_M = (1.0 \pm 0.2) \times 10^6$ M^{-1} s^{-1}, and hence $K_M = (10 \pm 2)$ nM. These values show that the optimization of the chain sequence of the end-grafted peptide was successful because it resulted in an Fc-peptide layer that displays a very high affinity for thrombin (low K_M). This high affinity, by compensating for the relatively slow kinetics of the cleavage step (low k_{cat}), results in an overall rapid, sensitive response of the modified Fc-peptide-bearing electrode to low thrombin concentrations.

(61) Page, M. J.; Macgillivray, R. T. A.; Di Cerra, E. *J. Thromb. Haemost.* **2005**, *3*, 2401–2408.
(62) Bode, W. *Blood Cells Mol. Dis.* **2006**, *36*, 122–130.
(63) Le Bonniec, B. F.; Myles, T.; Johnson, T.; Knight, C. G.; Tapparelli, C.; Stone, S. R. *Biochemistry* **1996**, *35*, 7114–7122.
(64) Backes, B. J.; Harris, J. L.; Leonetti, F.; Craik, C. S.; Ellman, J. A. *Nat. Biotechnol.* **2000**, *18*, 187–193.
(65) Su, Z.; Vinogradova, A.; Koutychenko, A.; Tolkatchev, D.; Ni, F. *Protein Eng. Des. Sel.* **2004**, *17*, 647–657.
(66) Petrassi, H. M.; Williams, J. A.; Li, J.; Tumanut, C.; Ek, J.; Nakai, T.; Masick, B.; Backes, B. J.; Harris, J. L. *Bioorg. Med. Chem. Lett.* **2005**, *15*, 3162–3166.
(67) Gosalia, D. N.; Denney, W. S.; Salisbury, C. M.; Ellman, J. A.; Diamond, S. L. *Biotechnol. Bioeng.* **2006**, *94*, 1099–1110.
(68) Senderoff, R. I.; Jiang, S. Stabilized Thrombin Compositions. Zymogenetics, Inc., Intellectual Property Department; U.S. Pat. Appl. 20080311104, **2008**.

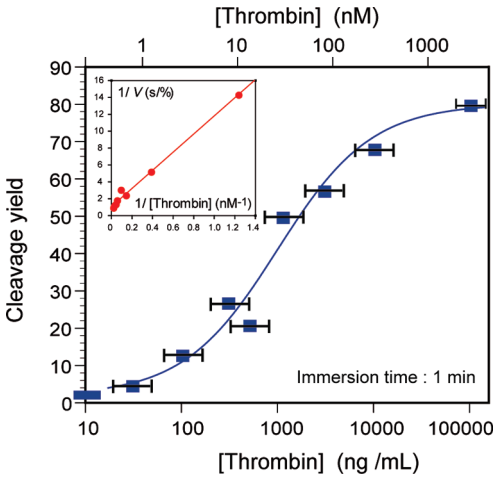


Figure 6. Electrochemical assay of proteolytic thrombin activity using an optimized Fc-[RFSRPQL]-PEG/MCH peptide layer. Shown is the calibration curve established for a 1 min immersion time of the sensing heptapeptide layer in active thrombin assay solutions at 37 °C. Error bars are the standard deviation of at least three repetitive experiments. The thick blue line crossing the cleavage yield axis is the point for [thrombin] = 0 nM. The solid blue line is meant to be a visual guide. The dynamic range of the redox heptapeptide layer covers physiologically relevant concentrations for thrombin proteolytic activity. The concentration of thrombin within the compositions of the present work can be varied, depending on the intended use, by adjusting the dilution volume. To control bleeding, the thrombin concentration will generally be from ~25 ng/mL to 5 μg/mL (~100 U/mL to ~20 000 U/mL).[68] (Inset) Kinetic analysis: Inverse plot of the initial velocity V of proteolytic cleavage as measured in terms of the variation of the cleavage yield over 60 s versus thrombin concentration. The solid line represents a fit to the surface Michaelis−Menten equation (eq 1). Horizontal error bars correspond to the uncertainty in the small volume of the viscous, glycerol-containing thrombin stock solution that had to be pipetted to prepare the dilute assay solutions.

So far, we have been using MCH as the backfilling molecule to space Fc-peptide molecules laterally. However, as mentioned above, the molecule that is to be used for backfilling should also ideally prevent the unwanted adsorption of protein molecules present in the fluid to be assayed. It has been shown that MCH molecules are moderately efficient in this role but that short PEG chains are able to prevent protein adsorption effectively.[69] This motivated us to assemble Fc-[RFSRPQL]-PEG layers backfilled by short PEG_6 molecules, starting from the corresponding disulfide molecule.

Effect of Using a Short PEG_6 Diluent Instead of MCH on the Cleavage Efficiency of Trypsin and Thrombin. Mixed Fc-[RFSRPQL]-PEG/PEG_6 monolayers were assembled on the surfaces of gold electrodes as described above in the case of MCH-backfilled redox peptide layers. The cyclic voltammogram recorded with an Fc-[RFSRPQL]-PEG/PEG_6-layer-bearing electrode at a high scan rate (20 V/s) is presented in Figure 7 (red trace).

Even though this signal is clearly a surface signal, because the current falls to the level of the background signal after the peak, it does not present the ideal characteristics of the CVs recorded at MCH-backfilled Fc-peptide electrodes in the same scan-rate range (cf. Figure 2). In particular, the peaks are broad and the peak-to-peak separation is large (~90 mV at 20 V/s). Moreover, the peak current is observed to be proportional to the scan rate v

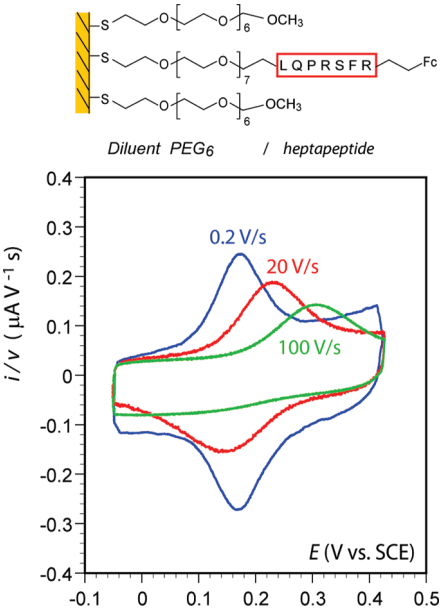


Figure 7. Cyclic voltammetry of a mixed Fc-[RFSRPQL]PEG/PEG_6 layer in a 1 M $NaClO_4$ aqueous solution at three scan rates: (blue trace) v = 0.2 V/s, (red trace) v = 20 V/s, and (green trace) v = 100 V/s. The current i is normalized vs v at T = 20 °C. The redox surface coverage, Γ_0, is ~8×10^{-12} mol/cm^2.

only at slow scan rates (v < 1 V/s for a peptide surface coverage of ~1×10^{-11} mol cm^{-2}); at higher scan rates, the ratio of the peak current over the scan rate decreases with increasing scan rate to an extent that is too large to be attributed to simple interference from the rate of electron transfer at the electrode (Figure S2 in Supporting Information). This behavior is characteristic of a surface-bound redox species whose access to the electrode proceeds at a finite rate (i.e., within a finite time that eventually becomes comparable to the characteristic observation rate of cyclic voltammetry, RT/Fv, at a very high scan rate).[70] We previously observed such behavior for redox-labeled short DNA chains end-grafted onto gold electrodes.[70] A thorough analysis of the variation of the signal characteristics with scan rate can yield information regarding the motional dynamics of the chain but is beyond the scope of the present work. We limit ourselves to underscore that the diffusion of the Fc head of the peptide to the electrode surface is significantly slowed when PEG_6 rather than MCH is used as a backfilling molecule. Such a slowing phenomenon may interfere with the electrochemical detection of protease activity because it gives high scan-rate signals a diffusive-like aspect, making the determination of the surface coverage in Fc-peptide chains inaccurate. Importantly, the longer the Fc-peptide chain or the longer and denser the backfilling layer, the more significant this effect may be. This phenomenon thus has to be taken into account in the design of protease-sensing redox peptide layers. Fortunately, by sufficiently lowering the scan rate, it is in principle possible to reach conditions where ample time is given to the Fc heads to reach the electrode so that chain dynamics does not participate in the kinetic control of the signal.[70] Accordingly, in the present case, the voltammograms recorded at a slow enough scan rate ($v \leq$ 0.2 V/s) were observed to display the characteristic features of ideal surface signals (Figure 7, blue trace), even though at such slow scan rates the contribution of the background current to the signal is markedly higher than at faster scan rates. These signals were used to monitor the activity of protease enzymes trypsin and thrombin.

(69) (a) Prime, K. L.; Whitesides, G. M. *J. Am. Chem. Soc.* **1993**, *115*, 10714–10721. (b) Ostuni, E.; Chapman, R. G.; Holmlin, R. E.; Takayama, S.; Whitesides, G. M. *Langmuir* **2001**, *17*, 5605–5620. (c) Chapman, R. G.; Ostuni, E.; Yan, L.; Whitesides, G. M. *Langmuir* **2000**, *16*, 6927–6936.

(70) Anne, A.; Demaille, C. *J. Am. Chem. Soc.* **2006**, *128*, 542–557.

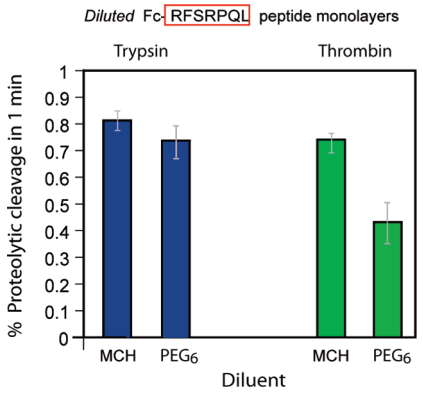


Figure 8. Proteolytic cleavage efficiency after 1 min of treatment at 37 °C with active trypsin (blue histograms) and active thrombin (green histograms) of Fc-[RFSRPQL]-PEG layers diluted with MCH or PEG_6. A comparison is made for similar redox surface coverages before immersion: $\Gamma_0 \approx 8 \times 10^{-12}$ mol/cm^2.

The proteolytic cleavage efficiency of an Fc-[RFSRPQL]-PEG/PEG_6 layer after 1 min of treatment with active trypsin is shown in Figure 8 and is compared to that achieved under similar conditions when MCH was used as a backfilling molecule (blue bars). Both efficiencies are similar and are around ~75%. It can thus be concluded that if trypsin is to be assayed then PEG_6-disulfide molecules can be used as surface diluents instead of MCH without compromising the ability of the enzyme to access and cleave the Fc-heptapeptide. However, such is not the case for thrombin because a 1 min immersion time in a 100 μg/mL thrombin solution results in only a 45% cleavage yield (vs a 75% cleavage yield when MCH was used a the diluent molecule), see green bars in Figure 8. The fact that the action of thrombin is hindered by the PEG diluent whereas trypsin action is not falls in line with the higher substrate specificity displayed by thrombin as compared to that displayed by trypsin.[52] These results also demonstrate that even though short PEG diluents are attractive candidates as backfilling molecules for Fc-peptide layers designed for protease sensing, because they oppose protein adsorption, the very same property is probably responsible for the limited response of the Fc-[RFSRPQL]-PEG/PEG_6 layer toward thrombin.

More generally, what is illustrated here is that, when designing protease-sensing peptide layers to be used in biological fluids, a delicate trade-off has to be found between giving the surface an overall protein-repelling nature, to prevent fouling, while preserving free access of the protease to the peptide substrate. The Fc-[RFSRPQL]-PEG/PEG_6 layer designed in the present work fulfills these seemingly incompatible requirements, at least for the detection of trypsin. Thrombin detection using the Fc-[RFSRPQL]-PEG/PEG_6 layer can probably be further improved, for example, by designing an Fc-peptide-PEG molecule composed of longer PEG and/or peptide chains.

Conclusions

The molecular structure of monolayers formed by Fc-peptide chains end-grafted onto electrode surfaces was shown to finely control the ability of these layers to sense the activity of serine protease enzymes trypsin and thrombin. The first determining structural parameter was the sequence and length of the Fc-peptide chain itself: optimal detection of trypsin could be achieved using a short Fc-[GPRS] tetrapeptide whereas a good response to thrombin was obtained only by using a longer Fc-[RFSRPQL] heptapeptide. This later peptide was also shown to be a good substrate for trypsin. The following general conclusions, valid for both the trypsin- and the thrombin-sensing electrodes and regarding the structure of the Fc-peptide layers, could be drawn. The surface coverage in Fc-peptide chains was shown to specifically limit the final yield of peptide cleavage. It was demonstrated that, even in diluted layers, a small fraction of the peptide chains appeared not to be cleavable by the enzymes. The cleavage yield was observed to increase when Fc-peptide chains were surface diluted using backfilling molecules such as MCH until the chains were sufficiently laterally spaced. Under these optimal conditions, the cleavage yield was independent of the chain surface coverage and was close to ~85%. Most remarkably, the response time of the assembled peptide layers was then shown to be exceptionally rapid (75% cleavage in 1 min). We attribute such a result to the fact that the Fc-peptide chains were linked to the electrode surface via a flexible PEG anchor designed to allow easy access to the cleavage site for the protease enzymes. It was observed that the cleavage of the Fc-peptide by the protease enzymes within these optimized diluted layers obeyed Michaelis−Menten kinetics. An analysis of the well-defined Lineweaver−Burk plot obtained permitted the full kinetic characterization of the Fc-peptide/enzyme systems (under simplifying assumptions), yielding values of both k_{cat} and K_M, which are the rate constants characterizing the kinetic behavior of the systems. The low K_M values obtained point to a high affinity of the two enzymes for their respective end-grafted peptide substrates whereas the cleavage step in itself was shown to be relatively slow. Overall, both parameters compensate and yield fast, responsive protease-sensing electrodes showing sufficient sensitivity to allow the detection of either trypsin or thrombin in the 1−1000 nM range. Finally, the molecule used to surface dilute (backfill) the Fc-peptide layers was shown to influence the response of Fc-heptapeptide bearing electrodes in various ways. Compared to the short MCH diluent molecule classically used to backfill end-grafted layers, a PEG_6 disulfide diluent was shown to slow the access of the peptide-borne Fc head to the electrode surface significantly. Interestingly, the reactivity of tryspin toward the end-anchored Fc-heptapeptide substrate seemed to be unaffected by the nature of the diluent. Conversely, a reduced cleavage yield was measured at thrombin-sensing electrodes backfilled with the PEG_6 diluent as compared to that measured at electrodes bearing MCH-diluted Fc-peptide layers. This result is important because when designing protease-sensing electrodes that are usable in biological fluids it might be necessary to use PEG rather than MCH diluent molecules, the former being more effective at preventing nonspecific protein adsorption than the later.

In agreement with the work presented here, a slowing effect of PEG chains on the interfacial kinetics of several proteases, including thrombin, has been very recently described in the case of peptide-capped gold nanoparticules.[71] However, the results that we obtained here for trypsin show that this effect is not universal but protease-dependent.

Supporting Information Available: Chemical structure of the Fc-peptide disulfides. Synthesis summary and characterization of the Fc-peptide PEG disulfides. HPLC proteolytic activity assays. CV characterization of the assembly of mixed Fc-peptide monolayers. CV characteristics of a mixed Fc-heptapeptide/PEG_6 monolayer. Detection of trypsin activity using a saturated undiluted Fc-tetrapeptide monolayer. Control experiments for the specificity of thrombin activity using Fc-heptapeptide/MCH layers. This material is available free of charge via the Internet at http://pubs.acs.org.

(71) Free, P.; Shaw, C. P.; Lévy, R. *Chem. Commun.* **2009**, 5009–5011.